

The notional category of evidentiality

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Abstract This chapter is a focused discussion of analytical options proposed for evidentiality in recent formal literature as well as the empirical motivation behind those options. Much work has been devoted to the ontological relationship between evidentiality and (epistemic) modality, and the chapter pays special attention to the types of data that motivate the analysis of evidentials as intensional operators. My ultimate goal is to sharpen the dialectic. I re-examine the tests used in existing literature and provide new empirical diagnostics that probe the intensionality of evidential utterances. My hope is for the reader to emerge with a better understanding of evidentiality as a semantic category—and with many directions for future work.

Keywords cross-linguistic semantics, evidentiality, modality, not-at-issue content, semantic theory, semantics/pragmatics division of labor

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And as the evidence, derived from witnesses and human testimony, is founded on past experience, so it varies with the experience, and is regarded either as a proof or a probability, according as the conjunction between any particular kind of report and any kind of object has been found to be constant or variable. There are a number of circumstances to be taken into consideration in all judgments of this kind; and the ultimate standard, by which we determine all disputes, that may arise concerning them, is always derived from experience and observation.

—*Of Miracles*, David Hume

1 Introduction

One of the fundamental questions in formal semantics is how to model meaning. Semantic categories are instrumental in addressing it: they allow us to group linguistic expressions into natural classes and to make testable predictions about their behavior. This chapter is on evidentiality, typically defined as linguistic encoding of the information source for an utterance (Chafe and Nichols 1986; de Haan 2013b). Inspired by Kratzer’s classic work on modality in natural language (1977; 1981; 1991; 2012), it aims at delineating the nature of evidentiality as a semantic, rather than morphosyntactic, category through a focused investigation of analytical options proposed for evidentials in recent formal literature and the empirical motivation behind those options.

Much work within the typological tradition has been devoted to the ontological relationship between evidentiality and (epistemic) modality (Aikhenvald 2004; van der Auwera and Plungian 1998; Bybee 1985; Palmer 1986): are they one category? different? distinct but overlapping ones? The formal-semantic literature raises related issues. According to the so-called modal view, evidentials can be assimilated to modal operators within Kratzerian semantics (Izvorski 1997). According to the so-called illocutionary view, evidentials in at least some languages instantiate a different category and should be analyzed as illocutionary force modifiers that directly interact with the structure of speech acts (Faller 2002; Murray 2014).¹ These views have been motivated by superficially different phenomena and have often been argued to track genuine semantic heterogeneity within evidentials. This chapter focuses specifically on the reasons to adopt or reject the modal view on evidentiality, which I will argue is complementary—rather than opposing—to the illocutionary one. My overarching goal is to examine how evidentials fit into the larger class of natural language phenomena dealing with knowledge and belief and whether they can be attributed the same core semantics as other modal expressions.

Current debates about the boundary between evidentiality and epistemic modality typically concentrate on (i) the (dis)similarities between specific evidentials and canonical epistemic modals (e.g., *might*, *must* and their counterparts in other languages), and on (ii) the merits of modelling evidentials within a version of Kratzerian possible-worlds semantics. However, it is important to recognize that different types of modals each have their unique properties (see 2.1 EPISTEMIC MODALITY, 2.2 PRIORITY MODALITY and 2.3 CIRCUMSTANTIAL AND ABILITY MODALS) and that Kratzer’s system isn’t always the best framework for some aspects of modal utterances, including their context update potential (1.4. DYNAMIC SEMANTICS FOR EPISTEMIC MODALITY), inference patterns (1.2 FROM MODAL LOGIC TO MODAL SEMANTICS), or graded nature (3.2 MODALITY, GRADABILITY AND PROBABILITY); see also 1.3 THE CONSTRUCTION OF MODAL DOMAINS. Furthermore, modality comes in various syntactic shapes and isn’t limited to modal auxiliaries in languages that have them (Arregui et al. 2017; 4.3 MODAL INDEFINITES, 4.4 MODALITY IN ADJECTIVES AND ADVERBS, 4.5 SUB-LEXICAL MODALITY). Therefore, the fact that some evidentials may not behave like Kratzerian epistemic modals is not a reason to reject a modal account for such evidentials wholesale.

For the purposes of this chapter I will understand modality as making reference to intensional domains², in line with the large body of work on modality beyond grammatical modals,

1. This view on evidentials echoes a long tradition that treats epistemic modals as categorically distinct from other modals, often argued to be illocutionary modifiers or comments on propositional content (Bybee et al. 1994; Halliday 1970; 2.1 EPISTEMIC MODALITY).

2. See 3.2 MODALITY, GRADABILITY AND PROBABILITY on alternatives to quantification and 1.2 FROM MODAL LOGIC TO MODAL SEMANTICS on alternatives to possible-worlds semantics for modals.

including conditionals (Kaufmann and Kaufmann 2015), predictive expressions (5.3 MODALITY AND THE FUTURE) and attitude predicates (6.3 MODALITY AND ATTITUDE VERBS). I argue that this view allows us to refine the dialectic surrounding the nature of evidentiality and to discern whether evidentials can be understood as belonging to a broad class of intensional operators. I discuss the range of tests that have been used to adjudicate between different theories in existing literature and bring to the fore novel empirical diagnostics that target intensionality. The emerging picture is that at least some evidentials are best understood as modals. They bear many similarities to speech and attitude predicates, and the accessibility relations they lexicalize provide an instrumental way of grouping them into linguistically useful categories.

§2 provides empirical background on evidentials. §3 outlines the modal view on evidential meaning, highlighting points of contrast with other approaches. §4 discusses ways of diagnosing the modality of evidentials. §5 concludes. My hope is for the reader to come out with a better understanding of evidentiality at the semantics-pragmatics interface—and with many directions for future work.

2 Empirical properties of evidentials

Just like English obligatorily expresses temporal notions through the category of tense, languages like Tariana (Arawakan: Brazil; Aikhenvald 2003) obligatorily express how the speaker acquired the prejacent (e.g., via vision or other senses, through inference or assumption, or via hearsay). Based on many genetically and geographically unrelated languages that, like Tariana, have evidential paradigms, Aikhenvald (2004) establishes that evidentiality in those languages is an independent grammatical category that isn't a species of modality. Aikhenvald's explicit goal is to separate genuine evidentials (which express information source as their primary function) from other categories that may have an evidential flavor (dubbed 'evidential strategies'). Furthermore, according to Aikhenvald (2004:4), "[l]inguistic evidentiality has nothing to do with [...] indicating one's belief. All evidentiality does is supply the information source".

This rigid view has been criticized from a functional-typological perspective. Even though many languages spoken in Europe express evidential notions through the tense-aspect or modal paradigms, it's not obvious why they shouldn't be included in the study of evidentiality on equal footing with Tariana (Diewald and Smirnova 2010), especially given that it's difficult to pinpoint the one primary function of multi-functional expressions (Boye 2010). As discussed in §1 above for modality, formal-semantic literature recognizes that modals may have a common semantic core despite morphosyntactic differences. Likewise, Bittner (2014) argues that temporality (semantic category) can receive a unified treatment in languages with or without tense (grammatical category).

Evidentiality is expressed via varied morphosyntactic means (de Haan 2013a). Extensive research within formal approaches to meaning demonstrates that these diverse expressions have much in common semantically (Murray 2017), despite the fact that some of them are 'only' evidential strategies in Aikhenvald's sense, e.g., aspectual morphology in Bulgarian (Izvorski 1997; Koev 2017; Simeonova 2020; Smirnova 2012), modal verbs in German (Ehrich 2001; Matthewson and Truckenbrodt 2017; Mortelmans 2000; Schenner 2008), high adverbials in English (Ernst 2009; Krawczyk 2012) or optional sentence-final particles in Japanese (Davis and Hara 2014; Hirayama 2020; Matsubara 2017; McCready and Ogata 2007).

The primary concern of this chapter is the core semantics of evidential expressions: what do they encode and what are the formal mechanisms behind it? While evidentials are typically defined as encoding only information source, they function as epistemic markers in larger discourses where they signal the speaker's authority or communicate shared knowledge (Bergqvist

and Grzech 2023; Grzech and Bergqvist 2025; Mushin 2013). It is important to understand which aspects of their meaning are responsible for these patterns and whether pragmatic information is hard-wired. Below I discuss a range of properties associated with evidentials (see also extensive discussion in Murray 2017, 2021; Schenner 2010b): information source (§2.1), the speaker’s attitude (§2.2), discourse status (§2.3), and perspectival anchoring (§2.4). Taken together, these properties demonstrate that evidentials are best treated as providing reasons to believe and grounds for claims, and not just as expressions of information source. For reasons of space, I do not discuss scope taking in detail (see Murray 2017:31-34). Evidentials always take wide scope with clause-mate negation (de Haan 1997; Korotkova 2020a), and vary with respect to other semantic operators, including conditionals (§2.3) and tense (Faller 2007); some of these properties can be explained by their syntactic position.

2.1. Source

The defining characteristic of evidentials is that they hard-wire information source for an utterance: how the speaker (in root declaratives, §2.4) acquired the prejacent. (1) lists information sources commonly found in the world’s languages. Evidential paradigms, like most grammatical categories (Aikhenvald 2004; Speas 2004), typically distinguish between two and four values, while other evidential systems often show more fine-grained distinctions.³

(1) Taxonomy of information sources (based on Willett 1988)

DIRECT	INDIRECT	
	INFERENCE	HEARSAY
• visual	• assumption	• secondhand
• auditory	• observable results	• thirdhand
• other sensory		• narrative

Three common evidential types are illustrated below: sensory *dug* in Standard Tibetan (Tibeto-Burman: Tibetan Plateau, China; Garrett 2001; Kalsang et al. 2013), German inferential *wohl* (Germanic: Austria/Germany/Switzerland; Zimmermann 2004; Eckardt 2020), and Cheyenne reportative⁴ (Algonquian: Montana, USA; Murray 2010, 2016). For each of those expressions (and other expressions traditionally described as evidentials), the source requirement is entailed and cannot be cancelled.⁵

(2) Tibetan ‘*dug*: speaker has evidence from senses (sight, smell, etc)

kha lag ’di bro ba chen po ’dug #yin na’i ngas bro ba bltas
 food this taste big POS EXIST.DIR but 1.ERG taste look.PST
 med
 NEG.EXIST.EGO

‘This food is tasty (I witnessed), #but I haven’t tasted it.’ (Muñoz 2019:5)

3. Some authors also include ‘ego(phoric) evidentiality’ (privileged knowledge about the prejacent). Part of the same paradigm as evidentiality in Tibeto-Burman languages, egophoricity is usually considered a separate category grammatically and conceptually (Floyd et al. 2018; Widmer 2019).

4. Cheyenne reportative doesn’t have a citation form: its morphological realization depends on subject agreement.

5. Glosses: 1,3 number, ACC accusative, AGR agreement, CNTR contrastive, CONJ conjectural evidential, COP copula, DEF definite, DET determiner, DIR direct evidential, EGO egophoric, ERG ergative, EUPH euphonic, EXIST existential, FUT future, GEN genitive, ILLA illative, INAN inanimate, IND indirect evidential, INDEF indefinite, INF infinitive, INFER inferential evidential, NEG negation, NOM nominative, POS positive degree, PL plural, PRES present, PST past, PTCL discourse particle, PTCP participle, REP reportative evidential, SG singular, SUP superlative, TOP topic.

- (3) German *wohl*: speaker infers based on body of information

Hein ist wohl auf See. #Ich sehe Hein dort auf der Cutty
Hein be.3SG.PRES CONJ at sea I see.1SG.PRES Hein there at DEF Cutty
Sark, die gerade ausläuft.

Sark that straight sail.3SG.PRES

‘Hein is at sea (I suppose). #I can see Hein right there sailing on the Cutty Sark.’

(adapted from Eckardt 2020:10)

- (4) Cheyenne reportative: speaker has access to a report (heard, read, etc)

É-hó’táaheva-sèstse Floyd #naa oha ná-sáa-néstó-he-∅

3-win-REP.3SG Floyd and CNTR 1-NEG-hear-AGR-DIR

‘Floyd won (I hear), #but I didn’t hear that.’

(Murray 2010:55)

The choice between different evidentials in each individual language is constrained by various pragmatic factors beyond the source requirement (Faller 2012; Krawczyk 2012; Murray 2010). It is an open question whether ‘information source’ is a semantic primitive (Krawczyk 2012), or how it is related to credence and reliability (McCready 2015). Take direct evidentiality. Typically associated with sensory information, for some expressions directness has been argued to encompass ‘best possible grounds’, the best type of evidence that can be available for a given proposition (Cuzco Quechua =*mi*, Faller 2002; Southern Aymara =*wa*, Martínez Vera 2024). This includes evidence from senses, but also encyclopedic knowledge or reliable reports (cf. Cariani 2022 on ideal grounding).

Likewise, indirect evidentiality raises many questions about the conceptual underpinnings of source restrictions. Thus, expressions often called conjectural evidentials often cover a broad spectrum (with the exclusion of eye-witness cases): they describe defeasible inferences from the speaker’s body of knowledge, including assumptions, observations, speculations, and hearsay (Eckardt 2020; Faller 2002; Murray 2010). Importantly, such evidentials don’t encode mere uncertainty: unlike probability operators (e.g., *likely*), conjectural evidentials are not licensed in scenarios where the only evidence is purely statistical, though the speaker may rely on statistical information as part of their reasoning chain.⁶ Under this definition, English *might* can be described as a conjectural evidential (von Fintel and Gillies 2010; Matthewson 2015).

In addition to source restrictions, direction of reasoning and/or causality plays a role in the meaning of many evidential expressions (Davis and Hara 2014; Krawczyk 2012; Takubo 2009). Some expressions are bi-directional (the ASSUMED type in Krawczyk’s (2012) terminology): St’át’imcets *k’a* (Matthewson et al. 2007:205-206), English *presumably* (Krawczyk 2012:50-51), Sm’algayax *sn* (Brown 2025), or English *must*. They are compatible both with inferences from effect to explanation⁷ (e.g., from observable results to their cause) and with inferences based on general assumptions or eliminative reasoning. Many expressions require the presence of an effect-to-explanation inference (the APPARENT type in Krawczyk’s (2012) terminology): St’át’imcets *an’* (Matthewson et al. 2007:205-206), Japanese *rashii* (McCready and Ogata 2007), Turkish indirect *mİş*, Central Alaskan Yup’ik *llini*, as well as English *evidently* and *apparently* (Krawczyk 2012). Conversely, predictive expressions, including English *will* or Italian evidential future, are incompatible with effect-to-explanation inferences (Winans 2016; Cumming and Winans 2021; Frana and Menéndez-Benito 2023).

Finally, several evidential expressions are sensitive to event structure, especially—but not

6. Epistemologists argue that purely statistical information isn’t sufficient for knowledge or assertibility, nor (possibly) for justified belief (Buchak 2014; Hawthorne 2004). I flag it here as a tentative argument in favor of strengthening the connection between evidentiality in language and justification as a basis for knowledge.

7. Such inferences are often called ‘abductive’ (Douven 2021).

exclusively—in those languages where they are part of the tense-aspect system, such as Tatar (see [Bowler 2018](#) for discussion and extensive typological overview). Such expressions have been argued to encode the time/place of evidence acquisition relative to the speaker’s here and now, with the overlap between those eventualities leading to ‘direct evidence’ and non-overlap leading to ‘indirect evidence’ ([Chung 2007](#); [Bowler 2018](#); [Faller 2004](#); [Koev 2017](#); [Nikolaeva 1999](#)). [Speas \(2010\)](#); [Kalsang et al. \(2013\)](#) propose that all information source restrictions can be reduced to underlying event ontology, however, this claim is too strong. Many evidentials hard-wire both evidence source information and the relative ordering of the evidence acquisition time and the prejacent time ([Hirayama and Matthewson 2022](#)). Independent of prejacent tense, this parameter of evidential meaning is similar to the notion of temporal perspective for modals ([Condoravdi 2002](#); 5.1 MODALITY AND TENSE).

Summing up, evidentials lexically encode information source for an utterance and may be additionally sensitive to other constraints.

2.2. Attitude

Evidentials systematically give rise to non-detachable inferences about the speaker’s attitude towards the prejacent, diagnosed by the impossibility of continuations that express disavowals and usually analyzed as entailments about belief or discourse commitment ([van Elswyk 2023](#); §3). All non-reportative evidentials require the speaker to consider the prejacent a live possibility (5), sometimes allowing weaker configurations $EV\phi \wedge EV\neg\phi$ (§4.2). Reportative evidentials often allow the speaker to void their belief ([AnderBois 2014](#)). Such evidentials are subject to cross-constructual variation with respect to disavowals ([Korotkova 2026](#)): (i) many encode nothing about the speaker’s stance and allow follow-ups of the form $\neg\phi$ (6),⁸ while (ii) some encode doxastic notions and are more constrained in the range of available follow-ups (7).

(5) German *wohl*

#Hein ist wohl auf See, und/aber er ist nicht auf See.
 Hein be.3SG.PRES CONJ at sea and/but he be.3SG.PRES NEG at sea
 Intended: ‘Hein is at sea (I suppose), and/but actually he isn’t at sea.’ ([Eckardt 2020:6](#))

(6) Cheyenne reportative

É-hó’táaheva-sêestse Floyd naa oha e-sáa-hó’táaheva-he- $\emptyset$
 3-win-REP.3SG Floyd and CNTR 3-NEG-win-AGR-DIR
 ‘Floyd won (I hear), but he didn’t (I find).’ ([Murray 2010:58](#))

(7) English *apparently*

Context: *Maria didn’t come to the party yesterday. Ana tells you that she was out of town. But you think that you might have seen her on the street.*

Apparently, Maria was out of town ✓but I don’t think it’s true/#but I believe that this isn’t the case/#but I know she wasn’t. ([Korotkova 2026](#))

[Forker \(2018\)](#) notes that those evidentials that ban disavowals likely instantiate an overlap between evidentiality and epistemic modality. However, based on extensive research within formal semantics, this is not an arbitrary property of some evidentials, but a hallmark of evidentiality as a category. The fact that different types of evidentials systematically give rise to inferences about the speaker’s doxastic state demonstrates that they encode more than informa-

8. The second clause in (6) features the morphologically unmarked direct evidential, which, as [Murray \(2016\)](#) notes, is difficult to adequately translate into English.

tion source for the prejacent (pace Aikhenvald).

2.3. At-issueness

Evidential meaning is typically not-at-issue, backgrounded information that is not part of the main point of an utterance (Murray 2010, 2014, 2017). Unlike canonical presuppositions and like many parenthetical constructions, evidentials contribute discourse-new information and are felicitous when used discourse-initially. In line with research on not-at-issue content more broadly (see overview in Koev 2018), the literature on evidentiality offers different ways of construing and diagnosing evidential not-at-issueness. For example, it is often argued that the evidential restriction is not-at-issue because it is not directly challengeable in conversation (*No, you haven't seen/heard/conjectured this* in an infelicitous reply to evidential utterances). However, the diagnostic applies only to some notions of at-issueness, and disagreement is independently infelicitous with first-person content, including evidentials (Korotkova 2016a,b). As discussed in detail in (Korotkova 2020a), the behavior of evidentials is best explained via the QUD notion of at-issueness according to which at-issue content is construed as QUD-relevant, and not-at-issue content as QUD-irrelevant (Beaver et al. 2017; Tonhauser et al. 2013, 2018). QUD at-issueness is diagnosed via answerhood, the ability to address explicit or implicit questions (Tonhauser 2012). Evidentials are naturally used when their prejacent answers the QUD (Context 1 in 8) and are usually infelicitous when the prejacent is QUD-irrelevant (Context 2 in 8).⁹

(8) ✓Context 1: Prejacent QUD-relevant, evidential signal QUD-irrelevant

You learn via the grapevine that your friends are organizing a surprise birthday party—for you. You hate surprises, and asked them more than once not to do that! When someone asks you why you are so upset, you say:

#Context 2: Prejacent QUD-irrelevant, evidential signal QUD-relevant

You learn via the grapevine that your friends are organizing a surprise birthday party—for you! You love surprises, and wish you hadn't heard about it. When someone asks you why you are so upset, you say:

German *sollen*

Meine Freunde **sollen** eine Party für mich organisieren. Echt schade.
my friend.PL REP.3PL.PRES INDEF party for me organize.INF total pity
'My friends are throwing a surprise party for me (I heard). Bummer.'

Evidentials are often compared to slifting parentheticals (*Voyager is coming home, I heard*) and are usually argued to be conventional markers of not-at-issue content (Faller 2019, 2023; Koev 2017; Martínez Vera 2024; Murray 2010, 2014, 2017).¹⁰ Korotkova (2020a, 2026) ar-

9. Evidentials have been argued to be QUD not-at-issue because they don't answer questions about evidence (cf. Bary and Maier 2021; Faller 2019; Korotkova 2020a; Lee 2013; Martínez Vera 2024):

- (i) A. How do you know that Jane is coming?
B. #Apparently, she is/✓I heard she is.

However, the prejacent of evidentials needs to contribute new information not settled by the immediate common ground (Korotkova 2026), so (i) may be infelicitous be due to constraints on givenness, rather than at-issueness. (8) doesn't run into this problem.

10. Evidential meaning is not a Pottsian conventional implicature (Murray 2010, 2014, 2017): evidentials, unlike Pottsian supplements (e.g., appositives and *as*-parentheticals), can influence the speaker's commitment to at-issue content.

gues instead that evidentials are conversationally not-at-issue (cf. Hunter 2016; Koev 2022 on parentheticals): the evidential restriction is QUD-irrelevant by default, but can become QUD-relevant in those discourses that warrant it. Evidentials vary in the availability of a non-parenthetical, locally accommodated reading in the scope of semantic operators, such as speech and attitude reports (§2.4) or conditionals: some can take narrow scope in conditionals ($\approx [[\text{EV}\phi] \rightarrow \psi]$; St’át’imcets, Matthewson et al. 2007; Japanese, McCready and Ogata 2007), some always project ($\approx \text{EV}\phi \wedge [\phi \rightarrow \psi]$; Cuzco Quechua =*si*, Faller 2002; German *wohl*, Eckardt 2020), and some exhibit variability (Tagalog *daw*; Kierstead 2015).¹¹ Importantly, it’s a consensus that evidentials contribute parenthetical meaning at the matrix level and are most naturally used to provide grounds for some QUD-relevant information conveyed by the prejacent.

2.4. Perspective

Evidentials belong to the heterogeneous group of ‘perspectival expressions’ (Bylinina et al. 2015; Mitchell 1986; Zu 2018): they reflect the viewpoint of a sentient individual determined contextually or, in some cases, structurally. Some expressions allow exocentric readings whereby the perspectival center is a contextually-salient third party (Barlew 2016; Lasersohn 2005; Partee 1989), and this is a possible distinction between evidentials and epistemic modal auxiliaries. *Might* has been claimed to have exocentric readings (Egan et al. 2005; 2.1 EPISTEMIC MODALITY). In contrast to that, evidentials are always speaker-oriented. (9) demonstrates that English *presumably* and German *wohl* cannot be anchored to a third party, even if this individual’s evidence for the prejacent is of the right type (see also §4.1).¹² Both expressions are felicitous in a similar scenario where they are speaker-oriented (10).¹³

(9) Context: Exocentric, third-party evidence

Ann is planning a surprise party for Bill, but Chris told Bill all about it. Now Bill and Chris are watching Ann set up the party without being discovered. Currently Ann is walking past Chris’ apartment carrying a large supply of party hats. She sees a bus on which Bill frequently rides home, so she jumps into some nearby bushes to avoid being spotted. Bill and Chris are both watching from Chris’s window and Chris asks Bill why Ann is hiding in the bushes. Bill says:

- a. ✓ I might be on that bus. (example and context from Egan et al. 2005)
- b. #I am presumably on that bus.
- c. German *wohl*
#Ich bin wohl auf diesem Bus.
I be.1SG.PRES CONJ at this bus
Intended: ‘I am on this bus (Ann supposes)’.

(10) Context: Autocentric, own evidence

Ann wants to avoid Bill and, seeing a bus on which Bill frequently rides home and

11. Some reportative evidentials can take wide scope in conditionals, yielding a speech report about the entire sentence $[\text{EV}[\phi \rightarrow \psi]]$ (Tagalog *daw*; Paraguayan Guaraní =*ndaje*, Tonhauser 2013). This is consistent with treating such reportatives semantically on a par with communicative predicates (13); the availability of this reading is likely enabled by syntax: all such evidentials are clitics and not part of the verbal spine.

12. *Presumably* belongs to what Ernst (2009) calls speaker-oriented adverbials, though Ernst doesn’t discuss exocentric contexts.

13. Korotkova (2016b) makes a stronger claim and argues that evidentials require a ‘de se’ construal.

someone resembling him boarding it, jumps into the bushes. Chris, watching from the window, asks her sotto voce what's going on. Ann says:

- a. ✓Presumably Bill is on that bus.
- b. German *wohl*
 ✓Bill ist wohl auf diesem Bus.
 Bill be.3SG.PRES CONJ at this bus
 'Bill is on this bus (I suppose).'

In other environments, evidentials often undergo perspectival shift. In ordinary information-seeking questions, evidentials exhibit ‘interrogative flip’ (term from [Tenny 2006](#)): they request the addressee to respond based on evidence specified by the evidential and never admit genuine speaker-oriented readings.¹⁴ In speech and attitude reports, evidentials often admit more than one interpretation ([Korotkova 2015](#); [Schenner 2010b,a](#); [Schwager 2010](#)): (i) a concord reading: evidential vacuously matches the matrix verb (e.g., reportative evidentials under communicative predicates); (ii) a parenthetical reading: evidential evaluated relative to a higher world (like syntactically embedded speaker-oriented adverbials); and (iii) a shifted, non-parenthetical reading: evidential evaluated relative to worlds introduced by the matrix verb (default interpretation of world-sensitive expressions in intensional environments, [Farkas 1997](#); [Percus 2000](#)).¹⁵ Importantly, not all evidentials have each reading, and it is clear that the semantics of the predicate at least partially determines which evidentials can occur in its scope and with which interpretation. Furthermore, evidentials in some languages do not appear in complement clauses due to independent syntactic constraints on clausal complementation ([Korotkova 2021](#)). It is an open issue how perspective shift with evidentials should be analyzed and whether it can be assimilated to other grammatical or pragmatic mechanisms ([Korotkova 2016b, 2020b](#); [Murray 2021](#)). I come back to the importance of a proper understanding of evidentials in root clauses in §4.1.

To recapitulate, evidentials—whether they are expressed via dedicated evidential paradigms, or other grammatical or lexical means—exhibit a range of cross-linguistic similarities that go beyond the semantic encoding of information source for an utterance. First, evidentials encode perspectival information and are anchored to the speaker in root declaratives. Second, all non-reportative evidentials and some reportative evidentials give rise to entailments about the speaker’s attitude. Third, in typical discourses only the prejacent proposition contributes the main point of an utterance, with the evidential restriction conveying not-at-issue, peripheral information.

14. [San Roque et al. \(2017\)](#) challenge the universality of the evidential interrogative flip. First, they don’t control for clause type: some of their examples are *wh*-exclamatives (an environment where evidentials are not expected to shift, [Bhadra](#) forth.). Second, most apparent exceptions come from non-canonical questions, an umbrella term for constructions that deviate from ordinary information-seeking interrogatives ([Eckardt et al. forth.](#)). [Korotkova \(2016b:253-255\)](#) proposes that evidentials may remain speaker-oriented only in those questions that don’t require the speaker’s ignorance (e.g., quiz, rhetorical or biased questions); see detailed discussion ([Brown 2025](#); [Eckardt 2020](#); [Faller 2023](#); [Littell et al. 2010](#)) on evidentials in conjectural questions, and ([Bhadra 2020](#); [Frana and Rawlins 2019, 2025](#)) on the interaction of evidentials with question bias.

15. The availability of different readings in intensional environments has been used as an argument for or against treating certain evidentials as modals ([Matthewson et al. 2007](#); [McCreedy and Ogata 2007](#); [Sauerland and Schenner 2007](#)). First, all world-sensitive expressions—whether they are themselves extensional (adjective *deciduous*) or intensional (adjective *probable*)—are expected by default to refer to the local worlds, unless a separate mechanism allows interpretation relative to a higher world (e.g., a ‘de re’ reading or projection). Second, this argument is framework-specific. If evidentials are analyzed as Kratzerian modals high on the clausal spine, they are indeed expect to shift ([Hacquard 2010](#)) and to not have parenthetical readings ([Schenner 2010b](#); [Schwager 2010](#)). However, if evidentials are treated akin to speech and attitude predicates with a variable at-issue status (§3), their behavior in intensional environments is independent of whether or not they have a modal semantics.

3 Theoretical accounts of evidentiality

Approaches to evidentiality within formal theories of meaning can be grouped into two families: **intensional** accounts and **extensional** accounts. Intensional accounts are usually referred to as the modal view on evidentiality. Extensional accounts include both (i) **illocutionary** approaches, wherein evidentials interact with the structure of speech acts, and (ii) **event-based** accounts, wherein evidential restrictions are a by-product of event ontology. This section outlines the core tenets of the modal view, highlighting important contrasts with other approaches.

Modal accounts of evidentiality treat evidentials as propositional operators within Kratzerian possible-worlds semantics for modality and parameterize the variation in the evidential domain using different force-flavor combinations, much like it's done for modal expressions at large, including attitude predicates (1.3 THE CONSTRUCTION OF MODAL DOMAINS—THE KRATZERIAN VIEW, 3.1 MODAL FORCE AND ITS REALIZATION ACROSS LANGUAGES, 6.3 MODALITY AND ATTITUDE VERBS). While some modals lexicalize neither force nor flavor (Steinert-Threlkeld et al. 2023), evidentials usually hard-wire at least flavor, which corresponds to the source restriction. The first modal approach to evidentiality was proposed by Pancheva (Izvorski 1997). She treats Bulgarian evidential perfect as a Kratzerian epistemic modal: a universal quantifier over the most normal worlds selected by the stereotypical ordering source from the worlds compatible with what is known at the world of evaluation, with an indirect evidence presupposition. Similar analyses exist for other languages (Bhadra 2022; Brown 2025; Eckardt 2020; Faller 2011; Garrett 2001; Matsubara 2017; Matthewson et al. 2007; McCready and Ogata 2007; Peterson 2010; Smirnova 2012).

This section gives the precis of the modal view within Hintikka semantics, using evidentials from §2.1 as exemplars. This schematic analysis preserves the spirit of existing modal accounts wherein evidentials are intensional quantifiers. At the same time, it doesn't run into framework-specific problems, some of which have been used as arguments against the modal view as a whole. First, it allows for explicit perspectival anchoring¹⁶: because classic Kratzerian semantics doesn't specify whose information state is being tracked, it may overgenerate for evidentials, some of which are more constrained in root declaratives than canonical modals (§2.4, §4.1). Note that for clause-taking verbs, Hintikka semantics reflects the syntax: their subject fills a thematic position of the predicate. The argument structure of evidentials is an open question that I do not discuss here (see Pancheva and Zubizarreta 2019; Simeonova 2020).

Second, this implementation allows more flexibility with respect to at-issueness and therefore has more purchase when it comes to the discourse dynamics of evidential utterances, including their behavior in conditionals (§2.3) and speech and attitude reports (§2.4). Most modal accounts implement the evidential restriction as a presupposition, however, this is not a defining property of the modal view. The system sketched below does not incorporate at-issueness explicitly. If one views evidentials as conventional triggers of not-at-issue content, they can be treated as speech and attitude predicates appearing in slifting parentheticals (*Voyager is coming home soon, I guess*), as in (Murray 2010, 2014, 2017).¹⁷ If one views evidentials as conversationally not-at-issue, they can be treated in parallel to 'discourse parentheticals' (*I guess Voyager is coming home soon*), constructions that have parenthetical meaning without parenthetical syntax (Hunter 2016; Simons 2007). In each case the key intuition is that evidentials

16. For simplicity, (11–13) fix the evidence holder to the speaker to account for root declaratives. These entries need to be modified to account for evidentials in other environments (§2.4).

17. Murray analyzes slifting parentheticals and Cheyenne evidentials on a par, arguing they lack a modal operator. However, verbs used in slifts do have intensional semantics. Therefore, even though evidentials can/must be used parenthetically, it doesn't mean that they lack modal semantics.

may be attributed a modal semantics, just like speech and attitude predicates, and that this is conceptually independent of the at-issue status of the evidential restriction (cf. also §4.1).

3.1. Direct evidentials

Direct evidentials require the speaker to have evidence from senses (and, in some cases, ‘best possible grounds’; §2.1), with similar restrictions lexicalized by English predicates *notice* and *observe*. Such evidentials are usually taken to encode full certainty (their compatibility with less-than-full certainty has not been tested, e.g., ‘I am not fully certain whether ϕ ’). Utterances with them are regarded as informationally stronger than unmodified assertions, which are omnivorous with respect to evidence type (Faller 2002; Martínez Vera 2024; Murray 2010). Note that perception does not guarantee belief. English *hear/see* have distinct epistemic profiles based on the complement type, and with bare infinitives they don’t require belief (Barwise 1981; Higginbotham 1983; Moulton 2009; see also Müller 2020 on German). (11) below ensures the presence of belief by hard-wiring it, which correctly predicts that disavowals are contradictory.¹⁸

(11) Tibetan *dug*’

- a. Let $\text{SENS}(Sp, w)$ be the set of worlds compatible with the sensory evidence the speaker has at w and $\text{DOX}(Sp, w)$ the set of worlds compatible with the speaker’s beliefs at w .
- b. $\llbracket \text{'dug'} \rrbracket^w = \lambda\phi. \forall w' \in \text{SENS}(Sp, w)[\phi(w')] \wedge \forall v' \in \text{DOX}(Sp, w)[\phi(v')]$.
- c. $\llbracket (2) \rrbracket^w = \forall w' \in \text{SENS}(Sp, w)[\iota x. \text{FOOD}(x, w') \wedge \text{TASTY}(x, w')] \wedge \forall v \in \text{DOX}(Sp, w)[\iota x. \text{FOOD}(x, v) \wedge \text{TASTY}(x, v)]$.
- d. Disavowals banned:
 $\forall v \in \text{DOX}(Sp, w)[\phi(v)] \wedge \forall v' \in \text{DOX}(Sp, w)[\neg\phi(v')]$.

The lexical entry above correctly predicts that direct evidentials are felicitous when the prejacent is known/believed to be true (including with follow-ups like ‘I know/believe ϕ ’). This property has been often used as a test for epistemic modality: since epistemic modals like *must* arguably are incompatible with certainty and/or knowledge (see §3.2), only non-modal evidentials can be used when the prejacent is known to be true (Faller 2002; Matthewson et al. 2007; Papafragou 2000). Importantly, attitude predicates that universally quantify over doxastic alternatives are felicitous precisely in such contexts (albeit the corresponding follow-ups might be redundant): *Based on observations, I concluded that there is water on Vulcan. I know it’s true*. So the felicity of an evidential expression when the prejacent is known/believed to be true only tests whether that expression requires the speaker’s information state to be unsettled about the prejacent, but not whether it is an (epistemic) modal.

3.2. Inference evidentials

Inference evidentials are often considered to be the meeting point of evidentiality and epistemic modality (van der Auwera and Plungian 1998) as in many ways such evidentials pattern like English *might* and *must*. For example, German *wohl* is most natural when the speaker draws a conclusion based on the body of information available to them: e.g., one can utter *Oma ist wohl einkaufen gegangen* ‘Granny went shopping (I suppose)’ if it’s Friday, her shopping

18. An alternative is to distinguish between perception of situations (non-epistemic) vs. of propositions (epistemic), as is done in the research on perception reports.

day, and her shopping bag and purse are missing (Eckardt 2020:10). Inferential reasoning is also encoded by attitude predicates, e.g., English *ascertain* and *figure out*. (12) is a notational version of Eckardt’s (2020; 2025) semantics for *wohl* that treats it as a stereotypical weak necessity modal (cf. Izvorski 1997 on Bulgarian). On the one hand, it bans flat-out disavowals (5), and is also incompatible with knowledge that ϕ (Zimmerman 2004; Eckardt 2020). On the other, it allows doubt: *Oma ist wohl einkaufen gegangen, aber ich bin mir nicht ganz sicher* ‘Granny went shopping (I suppose), but I’m not fully certain’ is non-contradictory. (12) predicts these facts by requiring that the speaker’s information state is unsettled.

(12) German *wohl*

- a. Let $\text{STEREO}(Sp, w)$ be the set of stereotypical worlds compatible with the speaker’s standards and expectations at w , ψ some contextually available information, and $\text{DOX}(Sp, w)$ the set of worlds compatible with the speaker’s beliefs at w .
- b. $\llbracket \text{wohl} \rrbracket^w = \exists \psi. \lambda \phi. \forall w' \in \text{STEREO}(Sp, w) [\psi(w') \rightarrow \phi(w')] \wedge \exists v \in \text{DOX}(Sp, w). [\phi(v)] \wedge \exists v' \in \text{DOX}(Sp, w) [\neg \phi(v')]$.
- c. $\llbracket (3) \rrbracket^w = \exists \psi. \forall w' \in \text{STEREO}(Sp, w) [\psi(w') \rightarrow \text{AT.SEA}(\text{HEIN}, w')] \wedge \exists v \in \text{DOX}(Sp, w). [\text{AT.SEA}(\text{HEIN}, v)] \wedge \exists v' \in \text{DOX}(Sp, w) [\neg \text{AT.SEA}(\text{HEIN}, v')]$.
- d. Disavowals banned:
 $\exists v \in \text{DOX}(Sp, w). [\phi(v)] \wedge \exists v' \in \text{DOX}(Sp, w) [\neg \phi(v')] \wedge \forall v' \in \text{DOX}(Sp, w) [\neg \phi(v')]$.

Inferential evidentials are often assumed to be weaker than unmodified assertions or assertions with direct evidentials (Faller 2002; Murray 2010; Smirnova 2012), however, weakness is tricky to establish and can be formalized in different ways. *Wohl* is weaker than an assertion because of its modal force (semantics): it is compatible with follow-ups that allow less-than-full certainty. Other inferential evidentials have been argued to be variable-force modals since they allow both weak and strong readings (Brown 2025; Matthewson et al. 2007; Peterson 2010). At the same time, there might be true necessity operators that encode indirectness/inference. Thus, because *must*-claims are often perceived as weaker than unmodified assertions, *must* has been argued to lack universal force (Karttunen 1972; Kratzer 1991). von Stechow and Gillies (2010, 2021) propose instead that *must* is strong, but semantically indirect, which explains perceived weakness without hard-wiring it. Other approaches to the same puzzle include a weak, indirect *must* (Lassiter 2016; Roberts 2023); a weak, non-evidential *must* (Goodhue 2017); and a strong, non-evidential *must* (Mandelkern 2019). These debates, and especially the data they are based on, can be useful in adjudicating between different approaches to the strength of evidential expressions; see also Degen et al. (2019) for an experimental paradigm.

3.3. Reportative evidentials

Reportative evidentials, like speech reports at large, quantify over the body of information available to the speaker and thus can be classified as information modals. Many such evidentials, including the Cheyenne reportative (6), don’t encode belief and allow the speaker to assert $\neg \phi$. (13) adapts Kaufmann and Kaufmann’s (2020) analysis of *according to* adverbials in English and is a notational variant of the standard Hintikka semantics for speech predicates (cf. Ehrlich (2001); Schenner (2008); Schwager (2010) on evidentials). The acquaintance relation below is underspecified, as many reportatives don’t constrain the report’s type (the speaker could have read it, heard it from someone, etc). Reportatives that are more constrained can be modelled via more specific acquaintance relations, in the same vein as the differences between *write* vs. *whisper* (cf. Grimshaw 2015 on SAY-by-means). The subject of communicative verbs is

often the report’s author (*announce, claim*) and evidentials are more similar to those whose subject is the recipient (*overhear, read*). Some reportatives allow/require overt specification of the original author (‘quotative evidentials’; Aikhenvald 2004:177-178), but whether they differ semantically has not been tested.

(13) Cheyenne reportative

- a. Let $\text{CONTENT}(x, w)$ be the propositional content associated with the source x in w and R an acquaintance relation.
- b. $\llbracket \text{Cheyenne reportative} \rrbracket^w = \exists x. \lambda \phi. \forall w' \in \text{CONTENT}(x, w). [\phi(w')]$
 $\wedge \exists R. R(Sp, w, \text{CONTENT}(x, w)).$
- c. $\llbracket 4 \rrbracket^w = \exists x. \lambda \phi. \forall w' \in \text{CONTENT}(x, w). [\text{WIN}(\text{FLOYD}, w')]$
 $\wedge \exists R. R(Sp, w, \text{CONTENT}(x, w)).$
- d. Disavowals allowed:
 $\forall v' \in \text{DOX}(Sp, w) [\neg \phi(v')].$

The possibility of disavowals has often been used as an argument against the epistemic modal analysis of evidentials (Faller 2002, 2019; Smirnova 2012), however, it only applies to specific implementations of the modal view. The possibility to assert $\neg \phi$ without a contradiction may diagnose the presence of a certain quantification domain (e.g., a realistic modal base; Kratzer 2012), but not the existence of a modal operator. (13) allows disavowals because the quantification domain of the modal needn’t include the actual world or the speaker’s belief worlds. Not all speech reports behave this way (7), which can be parameterized using the same general schema by adding doxastic restrictions to the semantics of some reportative evidentials, as well as communicatives like *be right* and *confirm* (Korotkova 2026).

3.4. Discussion

Let’s take stock. Distinctions found in the evidential domain closely resemble the patterns observed with speech and attitude reports, and the modal schema above uses Hintikka machinery to make these parallels explicit.¹⁹ Many clause-taking predicates lexicalize source restrictions and differ in force. Furthermore, the differences in disavowals between reportative vs. non-reportative evidentials mirror the differences between communicative vs. doxastic predicates (Korotkova 2026). Speech reports allow the subject to reject the prejacent (*✓Starfleet announced/claimed/declared that Voyager was coming home, but I knew it wasn’t true.*). Attitude reports, on the other hand, require their subject to consider the prejacent a live possibility (*#I believed/concluded/guessed that Voyager was coming home, but I knew it wasn’t true.*). Doxastic predicates differ in force: some encode full belief (*know, infer*), while some encode weaker notions (*doubt, suspect*). Summing up, modal accounts offer a powerful apparatus that parameterizes the behavior of evidential expressions using the tools often applied to modal expressions more generally. Below I briefly discuss two main contenders: illocutionary and event-based accounts.

Illocutionary accounts Illocutionary accounts of evidentiality treat evidentials as force modifiers that semantically interact with the structure of speech acts (see 7.1 SPEECH ACTS AND

19. This is not to say that evidentials don’t differ from clause-taking predicates. However, just like with modal auxiliaries vs. attitude predicates, the question is whether these differences can be derived from syntax while maintaining a unified semantics (6.3 MODALITY AND ATTITUDE VERBS).

MODALITY). Under this view, evidentials tap into mechanisms responsible for grounding and justification, and directly determine the speaker’s (in root declaratives) commitment to the prejacent, e.g., by manipulating the quality threshold for assertions (Davis et al. 2007). The first account of this type was proposed for Cuzco Quechua by Faller (2002), who argues that evidentials operate at the level of sincerity conditions for a speech act. Another prominent account is due to Murray (2010, 2014, 2017), who treats evidentials as hedges (Benton and van Elswyk 2020) which may weaken the speaker’s commitment to at-issue content: direct evidentials propose an update to ϕ (like assertions), inference evidentials propose an update to $\Box/\Diamond\phi$ (like shifting parentheticals), and reportative evidentials, which may require no commitment (6), propose a trivial update to $\phi \vee \neg\phi$ (7.1 SPEECH ACTS AND MODALITY). The cornerstone of most such accounts is that evidentials are associated with construction-specific discourse updates (Faller 2019, 2023; Martínez Vera 2024; Northrup 2014) and are therefore taken to be categorically different from modals, which necessarily have an intensional operator in their semantics and thus modify commitment indirectly.²⁰ Initial research suggested that all semantic variation within evidentiality stems from the distinction between modal vs. illocutionary evidentials (see Korotkova 2016b, 2021 for arguments against this view), for example, evidentials that take the widest scope with respect to semantic operators have been argued to be illocutionary. However, scope taking with modals is subject to a variety of constraints (6.1 MODALS, NEGATION, CONCORD, AND POLARITY), and epistemic modals often have the widest scope (Hacquard 2011), so this is not an argument against the modal view. As Murray (2021) emphasizes, the illocutionary view on evidential meaning is not incompatible with an intensional semantics for evidentials and merely doesn’t require it.

Event-based accounts Purely event-based accounts treat evidentials as extensional operators whose only function is to encode a relation between eventualities within the same world (see overview in Bowler 2018), e.g., Faller (2004) argues that Cuzco Quechua *-sqa* encodes spatio-temporal distance between the prejacent eventuality and the deictic center, leading to ‘indirectness’. An important prediction of such accounts is the impossibility of weak commitment, diagnosed via uncertainty follow-ups: even indirect evidentials are predicted to give rise to full commitment by virtue of being used in assertions (which commit the speaker to the prejacent; Marsili 2024; Pagin and Marsili 2025). The prediction is borne out for English *by now* adverbials (14): they create indirectness effects while having a purely temporal semantics, and don’t affect commitment (Altshuler and Michaelis 2020). Such data help determine whether a purely temporal semantics is sufficient to explain the behavior of putatively temporal evidential expressions (cf. also Hirayama and Matthewson 2022).

(14) #Voyager came home by now, but I’m not completely sure.

The defining characteristic of modal accounts is the presence of an intensional operator, while other ingredients, such as discourse updates and temporal components, can be incorporated into this basic semantics. Thus, there is a deep connection between temporality and modality (5.1 MODALITY AND TENSE), and some evidentials have been analyzed as relative tenses with a doxastic component (Lee 2013; Johnson 2022; Smirnova 2012). Likewise, different modal expressions can affect the speaker’s commitment to the prejacent. Therefore, in order to refute the modal view we need to show that evidential expressions are not intensional.

20. With the exception of (Krifka 2019), where both epistemic modals and evidentials are treated as commitment modifiers.

4 Probing for modality

This section sharpens the dialectic concerning the modal view on evidentiality, inspired by research on modal future (especially [Cariani 2021](#); [Klecha 2014](#)). Predictive expressions have been analyzed as purely temporal operators within tense logic (5.3 MODALITY AND THE FUTURE) as well as purely discourse-based commitment weakeners ([Frana and Menéndez-Benito 2023](#)). Across languages, they don't pattern like epistemic modals (e.g., conjunctions with epistemics, or scope with negation). Nevertheless, they induce intensional effects and are therefore best understood as modals. I advocate a similar approach to at least some evidentials that can be shown to require an intensional operator in their semantics, though they may not pattern like *might/must*: §4.1 discusses the non-diagnostic of (dis)agreement; §4.2 disentangles existential force from weak commitment; §4.3 explores standard intensional effects with evidentials; §4.4 proposes a new test based on the interaction of evidentials with subjective predicates.

4.1. Modality and (dis)agreement

Epistemic modal auxiliaries in English give rise to complex patterns of agreement and disagreement in dialogues ([Khoo 2015](#); [Knobe and Yalcin 2014](#)). Some of these data have been used in the literature on evidentiality to draw a line between those evidentials that behave like English epistemic modal auxiliaries and those that don't. Consider (15), based on ([von Fintel and Gillies 2008](#):83) and the game of Mastermind:

(15) St'át'imcets (Salish: British Columbia, Canada)

Imagine a game where someone places some different colored pegs behind a screen and the other person has to guess the colors and the order after getting some clues. After some rounds where I give my son some hints about the solution, he says:

wá7 k'a i tseqwts'qw-a
be INFER DET.PL red-EXIST
'There might be some reds'.

Possible responses include:

- | | |
|---|--|
| a. wenácw; wá7 k'a
true be INFER
'That's true. There might be.' | b. wenácw; wá7
true be
'That's true. There are.' |
|---|--|

([Matthewson et al. 2007](#):223)

[Matthewson et al. \(2007\)](#) and later literature interpret the data in the following way. Epistemic *might* makes both the prejacent ϕ and the modalized proposition $\Diamond\phi$ available for propositional anaphora, as the English translations of (15) illustrate. The expectation is that one can agree with $\Diamond\phi$ only if it is at-issue. Therefore, only those evidentials that have a modal operator in their semantics are expected to allow the response in (15a), mirroring the pattern for *might*-claims ([Faller 2019, 2023](#); [Murray 2014, 2021](#)).²¹ However, anaphoric potential is linked to at-issueness only in some theories (the so-called proposal at-issueness, [Koev 2018](#)). As discussed in §2.3, evidentials are best understood using the QUD notion of at-issueness, diagnosed via the ability to answer questions. [Snider \(2017\)](#) demonstrates that anaphoric potential is neither necessary nor sufficient for QUD at-issueness. To this end, the data in (15) show that some evidentials, like St'át'imcets *k'a*, make more than one proposition available

21. Disagreement has also been used as a diagnostic, but it creates additional complications ([Korotkova 2016a,b](#)).

for anaphora. Such data shouldn't be taken as an argument that those evidentials that allow only the prejacent to be targeted by anaphora (the response in (15b)) lack a modal component in their semantics altogether.²² For example, with first-person attitude reports (which have a modal semantics) anaphora only targets the prejacent, and not the mental state:

- (16) A. I assume/guess/suspect that there are two reds.
 B. ✓ That's right, there are.
 B'. # That's right, you do.

Moreover, there is an important caveat about the data in (15). Mastermind scenarios are often used to demonstrate that English epistemic modals in root declaratives aren't anchored to the speaker's body of evidence but may refer to public knowledge (DeRose 1991) and sometimes to a third party's information state (2.1 EPISTEMIC MODALITY). The speaker, who has access to the actual state of affairs, is supposed to give a reply based on mutually available information or their understanding of evidence available to someone else. Not all inferential expressions allow this reading (note that these examples are different from purely exocentric cases in (9)). Consider (17), which avoids the issues with propositional anaphora discussed above. Responses with *might* or *possibly* are acceptable, but English *presumably* and German *wohl* are infelicitous (unless B means them to refer to their own information state) and pattern like first-person attitude reports:

- (17) *A and B are playing Mastermind. A is the codebreaker and B is the codemaker, knowing exactly how many red pegs there are (two). B is keeping track of what A has figured out but doesn't want to give away the solution just yet.*
 A. There are two reds.
 B. ✓ There might be. / Possibly there are.
 B'. #I assume/guess/suspect that there are.
 B''. #Presumably there are.
 B'''. German *wohl*
 #Es gibt wohl zwei rote Stifte.
 it give.3SG.PRES CONJ two red peg.PL
 Intended: 'There are two red pegs (I suppose).'

Evidentials have been assumed to be always speaker-oriented in root declaratives (§2.4). It has not been tested systematically whether they allow readings tracking mutual evidence (*≈for all we know*). If, as the data suggest, there is a split between evidentials that are acceptable in Mastermind scenarios (St'át'imcets *k'a* and *an'*) and those that are not (English *presumably*; German *wohl*; Italian evidential future, Frana and Menéndez-Benito 2023:144), this is a genuine case of semantic variation in the evidential domain: (i) some evidentials pattern like *might* and can track mutual evidence (*≈for all we know*), and (ii) some evidentials pattern like first-person attitude reports and track individual evidence; cf. Tancredi's (2007) distinction between objective, knowledge-based vs. subjective, belief-based epistemics. The behavior of *might*-claims can be accounted for in a number of ways without postulating relativist semantics (Rudin 2025), which may or may not be applicable to evidentials. Importantly, the

22. The same applies to QUD at-issueness. If we show that only the prejacent can be at-issue, it doesn't mean that there is no modal operator. Additionally, evidentials are often analyzed as necessity modals, and it is difficult to show that $\Box\phi$ —and not ϕ —is at-issue even for *must* (Roberts 2023:1255).

fact that some evidentials are not licensed in scenarios requiring them to ascribe evidence to a third party, or to make a claim based on shared information, does not show that these evidentials do not have a modal semantics; it only shows that these evidentials differ from *might* and can be modals of a different kind (like first-person attitude reports). Finally, Mastermind cases can only be tested with a subset of evidentials. Given the guessing nature of the game, it excludes sensory and reportative evidentials.²³ Additionally, not all inferential evidentials are compatible with the type of reasoning the game requires, as some of them are not licensed in speculative contexts that often lack the effect-to-explanation component (§2.1). Summing up, while the data in (15)–(17) offer valuable insights into the semantics of evidentials, they do not adjudicate between intensional vs. extensional theories of evidential meaning.

4.2. Existential force vs. weak commitment

For many evidentials, it is often difficult to empirically tease apart the predictions of accounts assigning them (near-)universal modal force vs. strong discourse commitment. But existential force is easier to detect. The hallmark of existential modals is the felicity of $\Diamond\phi \wedge \Diamond\neg\phi$.²⁴ Consider the behavior of Cuzco Quechua *=chá*, typically analyzed as an all-purpose conjectural evidential:

(18) Cuzco Quechua *=chá*

Inés-chá llali-rqa-n. Pilar-taq-chá llalli-rqa-n.

Inés-CONJ win-PST-3 Pilar-CNTR-CONJ win-PST-3

‘Possibly Inés won. And possibly Pilar won.’

Comment: non-contradictory even if the two women participated in the same race, and only one can be the winner. (Faller 2002:187)

Faller (2002) analyzes *=chá* as a force modifier whose prejacent is $\Diamond\phi$. Faller (2023) removes the modal part and treats *=chá* only as a commitment weakener (cf. Murray (2014) on the Cheyenne conjectural). However, weak commitment is commitment. (18) is then supposed to convey that the speaker weakly commits to ϕ and also to $\neg\phi$, without comments about possible belief revision, which amounts to confusion. On the other hand, (18) is straightforward under a modal analysis. *Might*-modals minimally encode that the speaker’s information state is decided on the non-impossibility that ϕ (the speaker can’t exclude ϕ) and can be strengthened pragmatically (Rudin 2019). (18), and its English translation, simply signal that the speaker excludes neither ϕ nor $\neg\phi$; it is a face-saving way of communicating ‘I don’t know’ (Eckardt 2025:432-433). I conclude that Cuzco Quechua *=chá* has an existential operator in its semantics (in line with Faller 2002, 2011). It is a separate issue whether its discourse effects can be derived from its core semantics (as Rudin argues for *might*) or whether they should be encoded directly. The availability of the pattern in (18) supports the modal account of evidentials that give rise to it and therefore offers a tangible diagnostic for intensional evidentials.

4.3. Intensional domains

Drawing on the literature on intensional environments, this section considers three diagnostics that identify the presence of a modal operator: modal subordination, presupposition blocking

23. Reportative evidentials are usually infelicitous if the original author is one of the interlocutors (similarly to obviation effects; Schwager 2010). So if the codemaker gives hints by saying something, the codebreaker can’t convey this by using a reportative evidential (≈There are two reds, you said).

24. Thus, German *wohl* is not an existential modal—the following is contradictory: *#Ines hat wohl gewonnen. Und Pilar hat wohl gewonnen.* (≈Ines won (I suppose). And Pilar won (I suppose).)

and contradictions allowed due to a ‘de re’ interpretation.

Modal subordination Modal Subordination (MS) is a phenomenon whereby anaphora and other forms of presupposition satisfaction are often blocked in intensional environments unless in the scope of another intensional operator (Roberts 1996, 2021). It is triggered, e.g., by modal auxiliaries, conditionals or disjunction. Several evidentials have been argued to be modal based on this diagnostic (Faller 2017; McCready and Ogata 2007; Lee 2013; Smirnova 2012). (19) demonstrates for Japanese inferential *mitai*:

(19) Japanese *mitai*:

We are in a cabin in the woods, with no window, listening to a howling noise outside. We think that we are hearing a wolf, or wolves; but in fact the north wind is just strong tonight.

- a. [ookami-ga]_i kita mitai da
wolf-NOM came INFER COP.PRES
‘[A wolf/Some wolves]_i has/have come (it seems).’
- b. [yatsu(ra)-wa]_i totemo onaka-o sukaseteiru *(mitai da)
it(they)-TOP very stomach-ACC emptied INFER COP.PRES
‘[It/they]_i is/are very hungry *(it seems).’ (McCready and Ogata 2007:166)

This diagnostic is particularly useful for inferential evidentials that closely resemble epistemic modals in their discourse effects. MS is expected to be triggered by (i) intensional evidentials (such as Japanese *mitai*), but not by (ii) extensional evidentials that weaken commitment or encode temporality. Thus, *by now* adverbials do not give rise to MS (20).

(20) ✓By Friday, Magda submitted [a paper]_i. It_i was on modality.

However it’s difficult to test with direct evidentials: MS is tied to commitment and isn’t obligatory. Therefore, we may not see it with direct evidentials not because they aren’t modal, but because they require commitment. The same consideration applies to reportative evidentials: e.g., Cuzco Quechua =*si* has been claimed to be used almost veridically without overt disavowals (Faller 2019), which may be responsible for the absence of MS effects.

Presupposition blocking Speech and attitude reports can block the presuppositions of their prejacent from projecting to the matrix level because they create a space for local accommodation (Heim 1992 and much later work). Korotkova (2026) shows that several reportative evidentials pattern like communicative predicates and function as intervening operators that may block their prejacent’s presuppositions. (21) demonstrates for German *sollen* and the soft trigger *nicht mehr* ‘anymore’; a corresponding sentence without the evidential is infelicitous. This test is only applicable to those reportative evidentials that don’t require belief.

(21) German *sollen*

You read in a reputable newspaper that Switzerland stopped drilling in the Arctic.

Die Schweiz soll in der Arktis nicht mehr bohren. Aber hä? Die
DEF Sw. REP.3SG.PRES in DEF Arctic NEG more drill.INF but PTCL DEF
Schweiz hat doch gar nicht in der Arktis gebohrt.
Sw have.3SG.PRES PTCL PTCL NEG in DEF arctic drill.PTCP
‘Switzerland supposedly won’t drill in the Arctic anymore. But wait a minute,
Switzerland hasn’t drilled there at all.’ (Korotkova 2026)

‘De re’, ‘de dicto’ and contradictions Intensional environments give rise to ‘de re’/‘de dicto’ ambiguities and are known to ameliorate contradictions because they allow different parts of the prejacent to be evaluated with respect to different worlds: expressions in the main predicate position must be read ‘de dicto’ while expressions outside of main predicate position can be read ‘de re’ or ‘de dicto’ (Anand and Korotkova 2022; Farkas 1997; Keshet 2010; Musan 1997; Schwarz 2012). Korotkova (2026) shows that several reportative evidentials, including Turkish *mİş* (22),²⁵ pattern like communicative predicates in this respect and allow prejacentes that otherwise sound contradictory (this test only applies to those reportative evidentials that don’t require belief). Corresponding sentences without evidentials are infelicitous, unless with a special quotative intonation (which is an intensional environment; cf. Anand and Korotkova 2022:623).

(22) Turkish *mİş*

You read an article about a very dangerous disease, claimed to be caused by a fungus, while in fact you know that it is a virus.

Dünya-nın en ölümcül virüs-ü virüs değil-miş.

world-GEN SUP deadly virus-ACC virus NEG.COP-IND

≈ ‘I hear that the world’s deadliest virus is not a virus’.

(Korotkova 2026)

Faller (2002:256) notes the following contrast between reportative =*si* (analyzed as a force modifier) and -*sqa* (analyzed as a spatio-temporal extensional operator, Faller 2004): =*si* allows a ‘de dicto’ interpretation in its scope (23), while -*sqa* doesn’t.

(23) Cuzco Quechua =*si*

a. Estados Unidos-pa rey-ni-n-si Peru-man chayamu-nqa.

States United-GEN king-EUPH-3-REP Peru-ILLA arrive-3FUT

‘The king of the United States will come to Peru (I hear).’

b. Ichaqa mana chiqaq-chu. Estados Unidos-pi-qa mana-n rey kan-chu.

But not true-NEG. States United-LOC-TOP not-DIR king be-NEG

‘But this is not true. There is no king in the United States.’

(adapted from Faller 2002:256)

Faller does not explain the contrast, however, her analysis predicts that =*si* and -*sqa* would pattern together and block ‘de dicto’ Furthermore, based on the English translation the DP in question, *Estados Unidos-pa rey* ‘king of the United States’, can be a presupposition trigger, in which case we observe a difference in presupposition blocking between =*si* (blocking licensed) and -*sqa* (blocking not licensed), also not predicted by Faller’s analysis in which there is no intervening operator allowing local accommodation in the semantics of =*si*. I propose an alternative take: like a range of reportatives in other languages, =*si* is an intensional operator (cf. Faller 2011), while =*sqa* is truly extensional.²⁶

25. Turkish *mİş* is an indirect evidential compatible with abductive inferences and hearsay (Johanson 2000; Krawczyk 2012; Slobin and Akşu 1982). Some speakers treat even reportative uses as having a doxastic component, but for many younger speakers it allows disavowals (Şener 2011) and therefore licenses presupposition blocking and intensional contradictions, see also discussion in (Korotkova 2026).

26. (23) features the future marker -*nqa*, however, even if it is an intensional operator (like many predictive expressions), such modals do not license ‘de re’/‘de dicto’ ambiguities or presupposition blocking, so the effect in (23) is due to the presence of =*si*.

4.4. Interaction with subjective predicates

Subjective expressions, which describe matters of opinion (*delicious*, *lugubrious*), can have an exocentric reading in root declaratives (Lasersohn 2005). *The defense was not scary* can be uttered autocentrically (by the defending student) or exocentrically (by the proud supervisor). Many approaches postulate a special implicit argument *j*, the judge, in the semantic denotation of such predicates making them semantically distinct from ‘objective predicates’ (*deciduous*, *fermented*). Reportative evidentials—like speech reports at large—block an autocentric reading. Because (24) can only mean that the judge is not the speaker, Murray (2017, 2021) argues that the evidence holder is distinct from the judge and that the denotation of the Cheyenne reportative must include a judge shifter.

(24) Cheyenne reportative

É-pèehéva’é-nèse

3-good-REP.INAN.SG

‘It is good (I hear).’

(Murray 2017:46-47)

If judges are included in the semantics at all, they correlate with worlds of evaluation ($\llbracket \text{good} \rrbracket^{(j,w)} = \lambda x. \text{GOOD}(x, \langle j, w \rangle)$). If an expression shifts the judge, it has to shift the world of evaluation as well, and theories that divorce judges from worlds make incorrect predictions for subjective expressions in intensional environments (Anand and Korotkova 2022). Judge-free theories are unproblematic as they assimilate subjective predicates to ordinary intersective predicates ($\llbracket \text{good} \rrbracket^w = \lambda x. \text{GOOD}(x, w)$). Therefore, the fact that the Cheyenne reportative shifts the judge of a subjective predicate in its scope indicates that it is not just a force modifier (pace Murray) but also an intensional operator (that minimally shifts the index, as in (13)—and possibly the judges, if they are included in the denotation). Furthermore, a subset of subjective predicates trigger an acquaintance inference (25a; term from Ninan 2014): the requirement that the judge has some experience with the object in question (the nature of experience varies; Anand and Korotkova 2018; Cariani 2022; Muñoz 2019; Soria-Ruiz 2024). A range of expressions can block this inference, including *must*-modals (25b) and inferential evidentials (25c), but not direct evidentials (2).

(25) a. The tea was delicious, #but I never tasted it.

b. The tea must have been delicious, ✓but I never tasted it.

c. German *wohl* (based on Korotkova and Anand 2021:521)

The tea is freshly brewed and the color is just right.

Der Tee ist wohl lecker, ich habe ihn aber noch nicht
def tea be.3SG.PRES CONJ delicious I have.1SG.PRES but it yet NEG
probiert.

try.PTCP

‘The tea is delicious (I suppose) but I haven’t tried it yet.’

Some authors argue that the acquaintance inference can be blocked only by intensional expressions (Klecha 2014; Ninan 2020). Some argue that the phenomenon is restricted to intensional expressions with evidential restrictions that semantically interact with the acquaintance inference (Anand and Korotkova 2018; Muñoz 2019; Korotkova and Anand 2021). Crucially, the pattern in (25) strongly speaks in favor of an intensional analysis of acquaintance-blocking expressions.²⁷ Depending on the specific analysis, direct evidentials, which do not

27. There are also non-linguistic accounts of acquaintance that don’t take intensionality or evidentiality into ac-

block the acquaintance inference, can be treated as intensional (just not indirect; (11)) or as extensional.

This section offers several concrete diagnostics that support the inclusion of a modal operator in the denotation of evidential expressions: (i) the possibility of statements of the form $EV\phi \wedge EV\neg\phi$, which diagnose existential modal force (e.g., Cuzco Quechua =*chá*); (ii) the possibility of modal subordination, presupposition blocking and contradictory prejacent, which diagnose intensional environments (e.g., Cuzco Quechua =*si*, German *sollen*, Turkish *mlş*); (iii) judge-shifting and acquaintance-blocking with subjective expressions, which is difficult to explain on a non-modal analysis (e.g., Cheyenne reportative, German *wohl*). These tests have not been applied consistently across languages. However, preliminary data suggest that even evidentials in Cuzco Quechua and Cheyenne, whose behavior has been used to motivate the illocutionary view as a whole, come out as having an intensional semantics. These diagnostics are limited and only apply to evidentials that don't require full certainty/commitment. Direct evidentials are difficult: they can be treated as universal quantifiers with full belief, as force modifiers with full commitment, or as spatio-temporal modifiers with commitment supplied by assertion. One way to move forward will be to gauge the behavior of direct evidentials against some known properties of modals: (i) necessity modals don't distribute over disjunction ($\Box[\phi \vee \psi] \not\models \Box\phi \vee \Box\psi$), (ii) epistemic modals outscope strong quantifiers (**every>must/might*; von Fintel and Iatridou 2003), (iii) epistemic necessity modals are incompatible with preferential doxastics (**must* under *hope*; Anand and Hacquard 2013). I leave exploring those possibilities for future research.

5 Outlook

The debates on the proper semantics of evidentials in many ways resemble, on the one hand, the debates on epistemic modals as propositional operators vs. comments on propositional content, and, on the other hand, the debates on predictive expressions as temporal vs. modal operators. Current approaches to evidentiality have often been motivated by superficially different data and therefore focus on distinct features of evidential expressions. For example, some evidentials encode temporal constraints, which motivates the inclusion of the evidence acquisition time in their semantics (Hirayama and Matthewson 2022; Johnson 2022; Lee 2013). On the other hand, some evidentials have non-trivial common-ground management properties (Grzech 2020; Martínez Vera 2024), which suggests that they directly make reference to discourse structure (much like discourse particles). The overarching goals of this chapter are to elucidate conceptual issues surrounding evidential meaning, to identify consequential differences between the existing accounts and to ascertain whether they can be brought to a common denominator, focusing on arguments for or against treating evidentials as a type of modal expression. The defining feature of the modal view is the presence of an intensional operator in the semantic denotation of evidential expressions, with force-flavor combinations determining their specific linguistic profile. This chapter re-examines existing arguments against modal accounts and provides several empirical tools that diagnose evidential intensionality.

Extensional evidentials that don't affect commitment (cf. (14)) can be treated as event operators. Extensional evidentials that affect commitment can be assimilated to parenthetical expressions that encode illocutionary effects semantically (cf. Koev 2022), as Murray argues for Cheyenne. However, if evidentials are shown to be intensional, it is worth checking how much mileage we can get out of their basic semantics. Canonical epistemic modals modulate commit-

count, but they make incorrect empirical predictions (Anand and Korotkova 2018; Soria-Ruiz 2024; Ninan 2020).

ment to the prejacent, which may be analyzed via construction-specific discourse updates for epistemic claims or via deriving commitment indirectly from epistemic probabilities (Rudin 2019). Similar analytical options exist for evidentials, and deciding between them taps into larger questions about the relationship between discourse and grammar present in the literature on question bias, imperatives and clause typing at large: can the update potential of any given utterance be derived from its core semantics in conjunction with a single update mechanism and general pragmatic reasoning? This chapter focuses specifically on the semantic question. It shows that evidentials can be analyzed within the same framework as speech and attitude reports, further classified into semantically useful subtypes based on the accessibility relations they lexicalize (e.g., inferential or reportative evidentials), much like it's traditionally done for clause-taking predicates (e.g., the distinctive profile of communicative (Anand et al. 2017) or intention predicates (Grano 2017) is determined by their flavor). My hope is that a consistent application of the diagnostics discussed here will help adjudicate between different theories of evidential meaning, thus carving a way towards a better understanding of evidentiality as a semantic category.

References

- Aikhenvald, A. Y. (2003). Evidentiality in Tariana. In A. Y. Aikhenvald and R. M. W. Dixon (Eds.), *Studies in Evidentiality*, pp. 131–164. Amsterdam: John Benjamins.
- Aikhenvald, A. Y. (2004). *Evidentiality*. Oxford: Oxford University Press.
- Altshuler, D. and L. A. Michaelis (2020). By now: Change of state, epistemic modality and evidential inference. *Journal of Linguistics* 56(3), 515–539.
- Anand, P., J. Grimshaw, and V. Hacquard (2017). Sentence embedding predicates, factivity and subjects. In C. Condoravdi (Ed.), *Lauri Karttunen FestSchrift*. CSLI.
- Anand, P. and V. Hacquard (2013). Epistemics and attitudes. *Semantics and Pragmatics* 6(8), 1–59.
- Anand, P. and N. Korotkova (2018). Acquaintance content and obviation. In U. Sauerland and S. Solt (Eds.), *Sinn und Bedeutung* 22, Berlin, pp. 161–173. ZAS.
- Anand, P. and N. Korotkova (2022). How to theorize about subjective meaning: A lesson from ‘de re’. *Linguistics and Philosophy* 45(3), 619–681.
- AnderBois, S. (2014). On the exceptional status of reportative evidentials. In T. Snider, S. D’Antonio, and M. Weigand (Eds.), *Semantics and Linguistic Theory (SALT)* 24, pp. 234–254. LSA and CLC Publications.
- Arregui, A., A. Salanova, and M. L. Rivero (Eds.) (2017). *Modality Across Syntactic Categories*. Oxford Studies in Theoretical Linguistics. Oxford University Press.
- van der Auwera, J. and V. Plungian (1998). On modality’s semantic map. *Linguistic Typology* 2(1), 79–124.
- Barlew, J. (2016). Point of view and the semantics of spatial expressions in Mushunguli and English. *Journal of Semantics* 33(2), 215–267.
- Barwise, J. (1981). Scenes and other situations. *Journal of Philosophy* 78(7), 369–397.
- Bary, C. and E. Maier (2021). The landscape of speech reporting. *Semantics and Pragmatics* 14(8), 1–48.
- Beaver, D. I., C. Roberts, M. Simons, and J. Tonhauser (2017). Questions under discussion: Where information structure meets projective content. *Annual Review of Linguistics* 3(1), 265–284.
- Benton, M. and P. van Elswyk (2020). Hedged assertion. In S. Goldberg (Ed.), *The Oxford Handbook of Assertion*, pp. 244–263. Oxford: Oxford University Press.
- Bergqvist, H. and K. Grzech (2023). The role of pragmatics in the definition of evidentiality. *STUF - Language Typology and Universals* 76(1), 1–30.
- Bhadra, D. (2020). The semantics of evidentials in questions. *Journal of Semantics* 37(3), 367–423.
- Bhadra, D. (2022). Evidentiality: Unifying nominal and propositional domains. In D. Altshuler (Ed.), *Linguistics Meets Philosophy*, pp. 243–274. Cambridge: Cambridge University Press.
- Bhadra, D. (Forth.). Counterexpectational *wh*-questions with miratives. In *The Oxford Handbook of Noncanonical Questions*. Oxford.
- Bittner, M. (2014). *Temporality: Universals and Variation*. Explorations in semantics. Wiley-Blackwell.
- Bowler, M. L. (2018). *Aspect and evidentiality*. Ph. D. thesis, University of California, Los Angeles.
- Boye, K. (2010). Semantic maps and the identification of cross-linguistic generic categories: Evidentiality and its relation to epistemic modality. *Linguistic Discovery* 8(1), 4–22.
- Brown, C. (2025). Conjectural questions in Sm’algayax. *Folia Linguistica* 59(1), 7–28.
- Buchak, L. (2014). Belief, credence, and norms. *Philosophical Studies* 169(2), 285–311.
- Bybee, J., R. Perkins, and W. Pagliuca (1994). *The evolution of grammar: Tense, aspect and modality in the languages of the world*. The University of Chicago Press.
- Bybee, J. L. (1985). *Morphology: The Study of Relation Between Meaning and Form*. Amsterdam: John Benjamins.

- Bylinina, L., E. McCready, and Y. Sudo (2015). Notes on perspective-sensitivity. In P. Arkadiev, I. Kapitonov, Y. Lander, E. Rakhilina, and S. Tatevosov (Eds.), *Donum semanticum: Opera linguistica etlogica in honorem Barbarae Partee a discipulis amicisque Rossicis oblata*. Moscow: Yazyki slavianskoy kul'tury.
- Cariani, F. (2021). *The Modal Future: A Theory of Future-Directed Thought and Talk*. Cambridge: Cambridge University Press.
- Cariani, F. (2022). Future-past asymmetries, evidential grounding, and projection. In M. Degano, T. Roberts, G. Sbardolini, and M. Schouwstra (Eds.), *Proceedings of the 23rd Amsterdam Colloquium*, pp. 62–68. ILLC.
- Chafe, W. and J. Nichols (Eds.) (1986). *Evidentiality: The Linguistic Coding of Epistemology*. Norwood: Ablex Publishing Corporation.
- Chung, K.-S. (2007). Spatial deictic tense and evidentials in Korean. *Natural Language Semantics* 15(3), 187–219.
- Condoravdi, C. (2002). Temporal interpretation of modals: Modals for the present and for the past. In D. Beaver, S. Kaufmann, B. Z. Clark, and L. D. C. Martínez (Eds.), *The Construction of Meaning*, pp. 59–88. CSLI Publications.
- Cumming, S. and L. Winans (2021). Counterfactuals and abduction. *Ergo* 8(12).
- Davis, C. and Y. Hara (2014). Evidentiality as a causal relation: A case study from Japanese *youda*. In C. Pinõn (Ed.), *Empirical Issues in Syntax and Semantics 10*, pp. 179–196.
- Davis, C., C. Potts, and M. Speas (2007). The pragmatic values of evidential sentences. In T. Friedman and M. Gibson (Eds.), *Semantics and Linguistic Theory 17*, Ithaca, NY, pp. 71–88. Cornell University.
- de Haan, F. (1997). *The Interaction of Modality and Negation: A Typological Study*. New York: Garland Publishing.
- Degen, J., A. Trotzke, G. Scontras, E. Wittenberg, and N. D. Goodman (2019). Definitely, maybe: A new experimental paradigm for investigating the pragmatics of evidential devices across languages. *Journal of Pragmatics* 140, 33–48.
- DeRose, K. (1991). Epistemic possibilities. *Philosophical Review* 100, 581–605.
- Diewald, G. and E. Smirnova (2010). Introduction. Evidentiality in European languages: The lexical-grammatical distinction. In G. Diewald and E. Smirnova (Eds.), *Linguistic Realization of Evidentiality in European Languages*, pp. 1–14. Berlin/New York, NY: De Gruyter Mouton.
- Douven, I. (2021). Abduction. In E. N. Zalta (Ed.), *The Stanford Encyclopedia of Philosophy* (Summer 2021 ed.). Metaphysics Research Lab, Stanford University.
- Eckardt, R. (2020). Conjectural questions: The case of German verb-final *wohl* questions. *Semantics and Pragmatics* 13(9), 1–17.
- Eckardt, R. (2025). The landscape of inferential evidentiality: Towards a unified framework. In F. Longo and D. Panizza (Eds.), *Proceedings of Sinn Und Bedeutung* 29, pp. 421–436.
- Eckardt, R., G. Walkden, and N. Dehé (Eds.) (Forth.). *The Oxford Handbook of Noncanonical Questions*. Oxford: Oxford University Press.
- Egan, A., J. Hawthorne, and B. Weatherson (2005). Epistemic modals in context. In G. Preyer and G. Peter (Eds.), *Contextualism in Philosophy: Knowledge, Meaning and Truth*, pp. 131–169. Oxford: Oxford University Press.
- Ehrich, V. (2001). Was *nicht müssen* und *nicht können* (nicht) bedeuten können: Zum Skopus der Negation bei den Modalverben des Deutschen. In R. Müller and M. Reis (Eds.), *Modalität und Modalverben im Deutschen*, pp. 149–176. Hamburg: Buske.
- Ernst, T. (2009). Speaker-oriented adverbs. *Natural Language and Linguistic Theory* 27, 497–544.
- Faller, M. (2002). *Semantics and pragmatics of evidentials in Cuzco Quechua*. Ph. D. thesis, Stanford University.

- Faller, M. (2004). The deictic core of ‘non-experienced past’ in Cuzco Quechua. *Journal of Semantics* 21(1), 45–85.
- Faller, M. (2007). Evidentiality above and below speech acts. Ms., University of Manchester.
- Faller, M. (2011). A possible worlds semantics for Cuzco Quechua evidentials. In N. Li and D. Lutz (Eds.), *Proceedings of SALT 20*, pp. 660–683. CLC Publications.
- Faller, M. (2012). Evidential scalar implicatures. *Linguistics and Philosophy* 35, 285–312.
- Faller, M. (2017). Reportative evidentials and modal subordination. *Lingua* 186–187, 55–67.
- Faller, M. (2019). The discourse commitments of illocutionary reportatives. *Semantics and Pragmatics* 12(8), 1–46.
- Faller, M. (2023). Conjectural speech acts in Cuzco Quechua. *Journal of Pragmatics* 214, 144–163.
- Farkas, D. (1997). Evaluation indices and scope. In A. Szabolcsi (Ed.), *Ways of Scope Taking*, pp. 183–215. Dordrecht: Springer Netherlands.
- von Fintel, K. and A. S. Gillies (2008). CIA leaks. *Philosophical Review* 117(1), 77–98.
- von Fintel, K. and A. S. Gillies (2010). Must ... stay ... strong! *Natural Language Semantics* 18(4), 351–383.
- von Fintel, K. and A. S. Gillies (2021). Still going strong. *Natural Language Semantics* 29, 91–113.
- von Fintel, K. and S. Iatridou (2003). Epistemic containment. *Linguistic Inquiry* 34(2), 173–198.
- Floyd, S., E. Norcliffe, and L. S. Roque (Eds.) (2018). *Egophoricity*. Amsterdam: John Benjamins.
- Forker, D. (2018). Evidentiality and its relations with other verbal categories. In A. Aikhenvald (Ed.), *The Oxford Handbook of Evidentiality*, pp. 65–84. Oxford: Oxford University Press.
- Frana, I. and P. Menéndez-Benito (2023). The evidential future in Italian. *Natural Language Semantics* 31(2-3), 139–178.
- Frana, I. and K. Rawlins (2019). Attitudes in discourse: Italian polar questions and the particle *mica*. *Semantics and Pragmatics* 12(16), 1–55.
- Frana, I. and K. Rawlins (2025). Bias and evidential meaning. In E. Csipak, J. David, and M. Liu (Eds.), *A Festschrift in Honour of Regine Eckardt*, pp. 94–107. Berlin: Humboldt-Universität zu Berlin.
- Garrett, E. J. (2001). *Evidentiality and Assertion in Tibetan*. PhD dissertation, University of California, Los Angeles.
- Goodhue, D. (2017). Must ϕ is felicitous only if ϕ is not known. *Semantics and Pragmatics* 10(14), 1–27.
- Grano, T. (2017). Logic of intention reports. *Journal of Semantics* 34, 587–632.
- Grimshaw, J. (2015). The light verbs *Say* and *SAY*. In I. Toivonen, P. Csúri, and E. van der Zee (Eds.), *Structures in the Mind: Essays on Language, Music, and Cognition*, pp. 79–99. Cambridge: MIT Press.
- Grzech, K. (2020). Managing Common Ground with epistemic marking: ‘evidential’ markers in Upper Napo Kichwa and their functions in interaction. *Journal of Pragmatics* 168, 81–97.
- Grzech, K. and H. Bergqvist (Eds.) (2025). *Expanding the Boundaries of Epistemicity*. Berlin: De Gruyter.
- de Haan, F. (2013a). Coding of evidentiality. In M. S. Dryer and M. Haspelmath (Eds.), *The World Atlas of Language Structures Online*. Leipzig: Max Planck Institute for Evolutionary Anthropology.
- de Haan, F. (2013b). Semantic distinctions of evidentiality. In M. S. Dryer and M. Haspelmath (Eds.), *The World Atlas of Language Structures Online*. Leipzig: Max Planck Institute for Evolutionary Anthropology.
- Hacquard, V. (2010). On the event relativity of modal auxiliaries. *Natural Language Semantics* 18(1), 79–114.
- Hacquard, V. (2011). Modality. In K. von Stechow, C. Maienborn, and P. Portner (Eds.), *Semantics: An international handbook of natural language meaning*, Volume 2, pp. 1484–1515. Berlin: De Gruyter.

- Halliday, M. (1970). Functional diversity in language as seen from a consideration of modality and mood in English. *Foundations of Language* 6(3), 322–361.
- Hawthorne, J. (2004). *Knowledge and Lotteries*. Oxford: Oxford University Press.
- Heim, I. (1992). Presupposition projection and the semantics of attitude verbs. *Journal of Semantics* 9, 183–221.
- Higginbotham, J. (1983). The logic of perceptual reports: An extensional alternative to situation semantics. *The Journal of Philosophy* 80(2), 100–127.
- Hirayama, Y. (2020). Restriction on evidence in evidentiality: the part-whole relation between situations. *Proceedings of Sinn und Bedeutung* 24(1), 341–358.
- Hirayama, Y. and L. Matthewson (2022). Evidential-temporal interactions do not (always) come for free. *Journal of Pragmatics* 193, 173–188.
- Hunter, J. (2016). Reports in discourse. *Dialogue and Discourse* 7(4), 1–35.
- Izvorski, R. (1997). The present perfect as an epistemic modal. In A. Lawson (Ed.), *Semantics and Linguistic Theory (SALT)* 7, Ithaca, NY, pp. 222–239. LSA and CLC Publications.
- Johanson, L. (2000). Turkic indirectives. In L. Johanson and B. Utas (Eds.), *Evidentials: Turkic, Iranian and Neighbouring Languages*, pp. 61–88. Berlin / New York, NY: Mouton de Gruyter.
- Johnson, K. (2022). Time and evidence in the graded tense system of Mvskoke (Creek). *Natural Language Semantics* 30(2), 155–183.
- Kalsang, J. Garfield, M. Speas, and J. de Villiers (2013). Direct evidentials, case, tense and aspect in Tibetan: Evidence for a general theory of the semantics of evidential. *Natural Language and Linguistic Theory* 31(2), 517–561.
- Karttunen, L. (1972). *Possible and must*. In J. Kimball (Ed.), *Syntax and Semantics*, Volume 1, pp. 1–20. New York / London: Seminar Press.
- Kaufmann, M. and S. Kaufmann (2015). Conditionals and modality. In S. Lappin and C. Fox (Eds.), *The Handbook of Contemporary Semantic Theory*, pp. 237–270. Hoboken, NJ: Wiley Blackwell.
- Kaufmann, M. and S. Kaufmann (2020). Talking about sources. In M. Asatryan, Y. Song, and A. Whitmal (Eds.), *NELS 50: Proceedings of the Fiftieth Annual Meeting of the North East Linguistic Society*, Amherst, Massachusetts. GLSA.
- Keshet, E. (2010). Split intensionality: A new scope theory of *de re* and *de dicto*. *Linguistics and Philosophy* 33(4), 251–283.
- Khoo, J. (2015). Modal disagreements. *Inquiry* 58(5), 511–534.
- Kierstead, G. (2015). Projectivity and the Tagalog Reportative Evidential. MA Thesis, Ohio State University, Columbus.
- Klecha, P. (2014). Diagnosing modality in predictive expressions. *Journal of Semantics* 31(3), 443–455.
- Knobe, J. and S. Yalcin (2014). Epistemic modals and context: Experimental data. *Semantics and Pragmatics* 7(10), 10–21.
- Koev, T. (2017). Evidentiality, learning events and spatiotemporal distance: The view from Bulgarian. *Journal of Semantics* 34(1), 1–41.
- Koev, T. (2018). Notions of at-issueness. *Language and Linguistics Compass* 12(12), 1–16.
- Koev, T. (2022). *Parenthetical Meaning*. Oxford: Oxford University Press.
- Korotkova, N. (2015). Evidentials in attitudes: do’s and don’t’s. In E. Csipak and H. Zeijlstra (Eds.), *Sinn und Bedeutung* 19, pp. 340–357.
- Korotkova, N. (2016a). Disagreement with evidentials: A call for subjectivity. In J. Hunter, M. Simons, and M. Stone (Eds.), *JerSem: The 20th Workshop on the Semantics and Pragmatics of Dialogue*, pp. 65–75.
- Korotkova, N. (2016b). *Heterogeneity and Universality in the Evidential Domain*. Ph. D. thesis, University of California, Los Angeles.

- Korotkova, N. (2020a). Evidential meaning and (not-)at-issueness. *Semantics & Pragmatics* 13(4), 1–24.
- Korotkova, N. (2020b). Interrogative flip and indexical shift are distinct phenomena. *Snippets* 39, 3–5.
- Korotkova, N. (2021). The embedding puzzle: Constraints on evidentials in complement clauses. *Linguistic Inquiry* 52(1), 210–226.
- Korotkova, N. (2026). Hearsay and (non-)endorsement. Ms., Utrecht University.
- Korotkova, N. and P. Anand (2021). *Find, must* and conflicting evidence. In P. G. Grosz, L. Martí, H. Pearson, Y. Sudo, and S. Zobel (Eds.), *Proceedings of Sinn und Bedeutung* 25, pp. 515–532.
- Kratzer, A. (1977). What ‘Must’ and ‘Can’ must and can mean. *Linguistics and Philosophy* 1, 337–355.
- Kratzer, A. (1981). The notional category of modality. In H.-J. Eikmeyer and H. Rieser (Eds.), *Words, Worlds, and Contexts*, pp. 38–74. de Gruyter.
- Kratzer, A. (1991). Modality. In A. von Stechow and D. Wunderlich (Eds.), *Semantics: An International Handbook of Contemporary Research*, pp. 639–650. de Gruyter.
- Kratzer, A. (2012). *Modals and Conditionals*. Oxford: Oxford University Press.
- Krawczyk, E. A. (2012). *Inferred Propositions and the Expression of the Evidence Relation in Natural Language. Evidentiality in Central Alaskan Yup’ik Eskimo and English*. Ph. D. thesis, Georgetown University.
- Krifka, M. (2019). Layers of assertive clauses: Propositions, judgements, commitments, acts. In Y. Hartmann and A. Wöllstein (Eds.), *Propositionale Argumente im Sprachvergleich: Theorie und Empirie*. Tübingen: Gunter Narr Verlag.
- Lasersohn, P. (2005). Context dependence, disagreement, and predicates of personal taste. *Linguistics and Philosophy* 28(6), 643–686.
- Lassiter, D. (2016). *Must*, knowledge and (in)directness. *Natural Language Semantics* 24(2), 117–163.
- Lee, J. (2013). Temporal constraints on the meaning of evidentiality. *Natural Language Semantics* 21(1), 1–41.
- Littell, P., L. Matthewson, and T. Peterson (2010). On the semantics of conjectural questions. In T. Peterson and U. Sauerland (Eds.), *Evidence from evidentials*, pp. 89–104.
- Mandelkern, M. (2019). What ‘must’ adds. *Linguistics and Philosophy* 42(3), 225–266.
- Marsili, N. (2024). The definition of assertion: Commitment and truth. *Mind & Language* 39(4), 540–560.
- Martínez Vera, G. (2024). Direct evidentiality and discourse in Southern Aymara. *Natural Language Semantics* 32, 1–34.
- Matsubara, J. (2017). *The Semantics and Pragmatics of the Japanese Evidentials -Rashii, -Sooda, and -Yooda: an Experimental Investigation*. Ph. D. thesis, Northwestern University.
- Matthewson, L. (2015). Evidential restrictions on epistemic modals. In L. Alonso-Ovalle and P. Menendez-Benito (Eds.), *Epistemic Indefinites*, pp. 141–160. Oxford: Oxford University Press.
- Matthewson, L., H. Davis, and H. Rullman (2007). Evidentials as epistemic modals: Evidence from St’át’imcets. In J. van Craenenbroeck (Ed.), *Linguistic Variation Yearbook* 7, pp. 201–254. Amsterdam: John Benjamins.
- Matthewson, L. and H. Truckenbrodt (2017). Modal flavour/modal force interactions in German: Soll, sollte, muss and m’üsste. *Linguistische Berichte* 255, 259–312. .
- McCready, E. (2015). *Reliability in Pragmatics*. Oxford: Oxford University Press.
- McCready, E. and N. Ogata (2007). Evidentiality, modality and probability. *Linguistics and Philosophy* 30(2), 147–206.
- Mitchell, J. E. (1986). *The Formal Semantics of Point of View*. Ph. D. thesis, University of Massachusetts, Amherst.

- Mortelmans, T. (2000). On the ‘evidential’ nature of the ‘epistemic’ use of the German modals *müssen* and *sollen*. *Belgian Journal of Linguistics* 14, 131–148.
- Moulton, K. (2009). *Natural Selection and the Syntax of Clausal Complementation*. Ph. D. thesis, UMass, Amherst.
- Muñoz, P. (2019). *On tongues: The grammar of experiential evaluation*. Ph. D. thesis, University of Chicago.
- Müller, K. (2020). Perception verbs and finite complement clauses. In C. Pinón and L. Roussarie (Eds.), *Empirical Issues in Syntax and Semantics* 13, pp. 55–79.
- Muñoz, P. (2019). Acquaintance inferences as direct evidential effects. Talk presented at the workshop “Expressing evidence”, University of Konstanz, Germany, June 6-8, 2019.
- Murray, S. E. (2010). *Evidentiality and the Structure of Speech Acts*. Ph. D. thesis, Rutgers University, New Brunswick, NJ.
- Murray, S. E. (2014). Varieties of update. *Semantics and Pragmatics* 7(2), 1–53.
- Murray, S. E. (2016). Evidentiality and illocutionary mood in Cheyenne. *International Journal of American Linguistics* 82(4), 487–517.
- Murray, S. E. (2017). *The Semantics of Evidentials*. Oxford: Oxford University Press.
- Murray, S. E. (2021). Evidentiality, modality, and speech acts. *Annual Review of Linguistics* 7(1), 213–233.
- Musan, R. (1997). *On the Temporal Interpretation of Noun Phrases*. New York: Garland.
- Mushin, I. (2013). Making knowledge visible in discourse: Implications for the study of linguistic evidentiality. *Discourse Studies* 15(5), 627–645.
- Nikolaeva, I. (1999). The semantics of Northern Ostyak evidentials. *Journal de la Société Finno Ougrienne* 88, 131–159.
- Ninan, D. (2014). Taste predicates and the acquaintance inference. In T. Snider, S. D’Antonio, and M. Weigand (Eds.), *Semantics and Linguistic Theory* 24, pp. 290–309. LS.
- Ninan, D. (2020). The projection problem for predicates of personal taste. In J. Rhyne, K. Lamp, N. Dreier, and C. Kwon (Eds.), *Semantics and Linguistic Theory* 30, pp. 753–778. LSA.
- Northrup, O. (2014). *Grounds for commitment*. Ph. D. thesis, University of California, Santa Cruz.
- Pagin, P. and N. Marsili (2025). Assertion. In E. N. Zalta and U. Nodelman (Eds.), *The Stanford Encyclopedia of Philosophy* (Fall 2025 ed.). Metaphysics Research Lab, Stanford University.
- Palmer, F. (1986). *Mood and Modality*. Cambridge: Cambridge University Press.
- Pancheva, R. and M. L. Zubizarreta (2019). On the role of person features in the evidential-temporal connection. *The Canadian Journal of Linguistics* 64(4), 673–708.
- Papafragou, A. (2000). *Modality: Issues in the semantics-pragmatics interface*. Amsterdam: Elsevier.
- Partee, B. H. (1989). Binding implicit variables in quantified contexts. In C. Wiltshire, B. Music, and R. Graczyk (Eds.), *Papers from the 25th Regional Meeting of the Chicago Linguistic Society*, pp. 342–365.
- Percus, O. (2000). Constraints on some other variables in syntax. *Natural Language Semantics* 8(3), 173–229.
- Peterson, T. (2010). *Epistemic Modality and Evidentiality in Gitksan at the Semantics-Pragmatics Interface*. Ph. D. thesis, University of British Columbia.
- Potts, C. (2005). *The Logic of Conventional Implicatures*. Oxford: Oxford University Press.
- Roberts, C. (1996). Anaphora in intensional contexts. In S. Lappin (Ed.), *Handbook of Contemporary Semantics*, pp. 215–246. Blackwell.
- Roberts, C. (2021). Modal subordination: “It would eat you first!”. In D. Gutzmann, L. Matthewson, C. Meier, H. Rullmann, and T. E. Zimmerman (Eds.), *The Wiley Blackwell Companion to Semantics*, pp. 1–36. John Wiley & Sons.

- Roberts, C. (2023). The indexical character of epistemic modality. *Linguistics and Philosophy* 46(5), 1219–1267.
- Rudin, D. (2019). Deriving a variable-strength *Might*. In N. Bade, P. Berezovskaya, and A. Schöller (Eds.), *Proceedings of Sinn Und Bedeutung* 20, pp. 587–603. Open Journal Systems.
- Rudin, D. (2025). Asserting epistemic modals. *Linguistics and Philosophy* 48(1), 43–88.
- San Roque, L., S. Floyd, and E. Norcliffe (2017). Evidentiality and interrogativity. *Lingua* 186–187, 120–143.
- Sauerland, U. and M. Schenner (2007). Embedded evidentials in Bulgarian. In E. Puig-Waldmüller (Ed.), *Proceedings of Sinn und Bedeutung* 11, pp. 495–509.
- Schenner, M. (2008). Double face evidentials in German: Reportative ‘sollen’ and ‘wollen’ in embedded contexts. In A. Grønn (Ed.), *Proceedings of Sinn und Bedeutung* 12, pp. 552–566. Open Journal Systems.
- Schenner, M. (2010a). Embedded evidentials in German. In G. Diewald and E. Smirnova (Eds.), *Linguistic realization of evidentiality in European languages*, pp. 157–185. Mouton de Gruyter.
- Schenner, M. (2010b). Evidentials in complex sentences: Foundational issues and data from German and Turkish. In T. Peterson and U. Sauerland (Eds.), *Evidence from Evidentials*, pp. 183–220. Vancouver: University of British Columbia.
- Schwager, M. (2010). On what has been said in Tagalog: Reportative *daw*. In T. Peterson and U. Sauerland (Eds.), *Evidence from Evidentials*, pp. 221–246. Vancouver: University of British Columbia.
- Schwarz, F. (2012). Situation pronouns in determiner phrase. *Natural Language Semantics* 20(4), 431–475.
- Şener, N. (2011). *Semantics and Pragmatics of Evidentials in Turkish*. PhD dissertation, University of Connecticut, Storrs.
- Simeonova, V. (2020). *The Syntax and Semantics of Light Attitudes*. Ph. D. thesis, University of Ottawa.
- Simons, M. (2007). Observations on embedding verbs, evidentiality, and presupposition. *Lingua* 117(6), 1034–1056.
- Slobin, D. and A. Akşu (1982). Tense, aspect, and modality in the use of the Turkish evidentials. In P. Hopper (Ed.), *Tense-Aspect: Between Semantics and Pragmatics*, pp. 185–200. Benjamins.
- Smirnova, A. (2012). Evidentiality in Bulgarian: Temporality, epistemic modality, and information source. *Journal of Semantics* 30, 479–532.
- Snider, T. (2017). *Anaphoric Reference to Propositions*. Ph. D. thesis, Cornell University.
- Soria-Ruiz, A. (2024). Expressive inferences and mighty obstacles. In F. Carcassi, T. Johnson, S. B. Knudstorp, S. D. Parrado, P. R. Robledo, and G. Sbardolini (Eds.), *Proceedings of the 24th Amsterdam Colloquium*, Amsterdam, pp. 358–364. ILLC.
- Speas, M. (2004). Evidentiality, logophoricity and the syntactic representation of pragmatic features. *Lingua* 114(3), 255–276.
- Speas, M. (2010). Evidentials as generalized functional heads. In A. M. DiSciullo and V. Hill (Eds.), *Edges, Heads and Projections: Interface Properties*, pp. 127–150. Amsterdam: John Benjamins.
- Steinert-Threlkeld, S., N. Imel, and Q. Guo (2023). A semantic universal for modality. *Semantics and Pragmatics* 16(1), 1–20.
- Takubo, Y. (2009). Conditional modality: Two types of modal auxiliaries in Japanese. In B. Pizziconi and M. Kizu (Eds.), *Japanese Modality*, pp. 150–182. London: Palgrave Macmillan.
- Tancredi, C. (2007). A multi-model modal theory of I-semantics. Ms., University of Tokyo <https://www.semanticsarchive.net/Archive/WFkYjU1Y/Multi-Model%20Modal%20Theory.pdf>.
- Tenny, C. (2006). Evidentiality, experiencers and the syntax of sentience in Japanese. *Journal of East Asian Linguistics* 15, 245–288.

- Tonhauser, J. (2012). Diagnosing (not-)at-issue content. In E. Bogal-Allbritten (Ed.), *Proceedings of the Sixth Conference on the Semantics of Under-Represented Languages in the Americas (SULA)*, Amherst, MA, pp. 239–254. GLSA Publications.
- Tonhauser, J. (2013). Reportative evidentiality in Paraguayan Guaraní. In H. Greene (Ed.), *Proceedings of the Seventh Conference on the Semantics of Under-Represented Languages in the Americas (SULA)*, Amherst, MA, pp. 189–204. GLSA Publications.
- Tonhauser, J., D. Beaver, C. Roberts, and M. Simons (2013). Towards a taxonomy of projective content. *Language* 89(1), 66–109.
- Tonhauser, J., D. I. Beaver, and J. Degen (2018). How projective is projective content? Gradiance in projectivity and at-issueness. *Journal of Semantics* 35(3), 495–542.
- van Elswyk, P. (2023.). Expressing belief with evidentials: A case study with Cuzco Quechua on the dispensability of illocutionary explanation. *Journal of Pragmatics* 203, 52–67.
- Widmer, M. (2019). Same same but different: On the relationship between egophoricity and evidentiality. In H. Bergqvist and S. Kittilä (Eds.), *Evidentiality, egophoricity, and engagement*, pp. 263–287. Berlin: Language Science Press.
- Willett, T. (1988). A cross-linguistic survey of the grammaticization of evidentiality. *Studies in Language* 12(1), 51–97.
- Winans, L. (2016). *Inferences of “Will”*. Ph. D. thesis, University of California, Los Angeles.
- Zimmerman, M. (2004). Zum Wohl: Diskurspartikeln als Satztypmodifikatoren. *Linguistische Berichte* 199, 1–35.
- Zu, V. (2018). *Discourse Participants and the Structural Representation of the Context*. PhD dissertation, New York University, NY.